From Pseudoclasses to Real Changes: Spatial–Spectral Difference Learning for Mitigating Spectral Pseudochange in Hyperspectral Change Detection

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***Abstract*—A core challenge in hyperspectral change detection (HCD) is spectral pseudochange, artifacts from imaging conditions that cause severe overlap between changed and unchanged samples, reducing their separability and hindering accurate detection. Existing methods, including traditional difference-driven pipelines and deep learning models, remain vulnerable to such discrepancies. To address this issue, we use pseudoclass separability as a design perspective to guide spatial–spectral representation enhancement, where changed and unchanged labels are regarded as discriminative surrogate categories defined on bitemporal spectral pairs rather than semantic land-cover categories. We propose spatial–spectral difference learning (SSDL), a spatial–spectral difference enhancement framework whose central component is spectral attention (SpeA) for peak-valley spectral-structure enhancement. SpeA is complemented by spatial differential attention (SDA) for competitive suppression of nonchange responses and position-aware multihead spectral linear self-attention (SpeLA) for spectral correlation modeling with a favorable accuracy-complexity tradeoff. Experimental results on three public HCD datasets demonstrate competitive and consistent performance, particularly in scenarios where spectral pseudochange causes strong changed/unchanged distribution overlap, and the code is available at https://github.com/WUTCM-Lab/SSDL**

***Index Terms*—Differential feature enhancement, hyperspectral change detection (HCD), spatial–spectral feature extraction, spectral attention (SpeA).**

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I. Introduction

HYPERSPECTRAL change detection (HCD) identifies land cover changes by analyzing spatial–spectral feature differences between multitemporal hyperspectral images. This task is widely applied in fields such as vegetation degradation, urban expansion, and disaster damage assessment, making it a core task in remote sensing [1], [2], [3]. On the one hand, hyperspectral images contain hundreds of contiguous spectral bands (with nanometer-level spectral resolution) and fine spatial texture information, both of which need to be modeled simultaneously to accurately capture changes [4]. On the other hand, differences in imaging conditions across temporal phases (e.g., light intensity, atmospheric scattering, and sensor noise) cause abnormal fluctuations in spectral reflectance over unchanged areas (i.e., "spectral pseudochanges"), significantly increasing the difficulty of distinguishing between changed and unchanged regions.

A core challenge in HCD lies in addressing spectral pseudochange, a phenomenon referring to apparent spectral discrepancies observed over unchanged surfaces between two temporal images. Unlike actual physical changes in the observed scene, these discrepancies stem from factors such as illumination variation, atmospheric effects, sensor noise, and radiometric or calibration mismatches. Spectral pseudochange typically manifests in two primary forms: unchanged pixels show significant fluctuations within specific bands and inconsistent reflectance at the same band positions across temporal phases, resembling the patterns in changed regions.

As visualized in Fig. 1, the average spectral curves of unchanged pixels exhibit significant variations within certain

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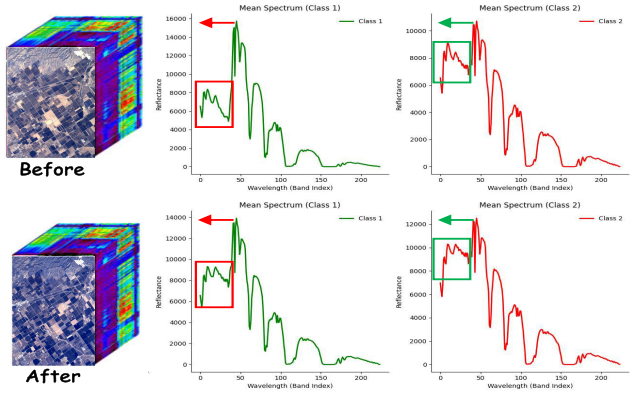


Fig. 1. Average spectral curves of changed and unchanged pixels in multitemporal hyperspectral images. Unchanged pixels exhibit significant spectral fluctuations in certain bands (marked by red rectangular boxes) and reflectance differences across different temporal phases in the same band (indicated by red arrows), a phenomenon similar to the green rectangular boxes and green arrows of pixels in changed regions.

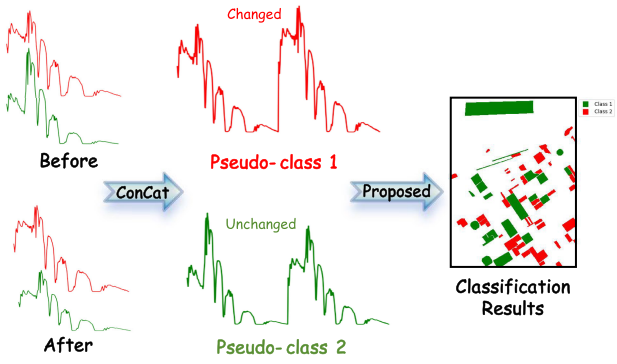


Fig. 2. Workflow of the proposed change detection framework. By concatenating multitemporal hyperspectral images along the spectral dimension (integrating prechange and postchange spectral information), labeling changed/unchanged pixels with pseudoclasses. The term "pseudo" emphasizes that the classes are not semantic land-cover categories but training surrogates defined on bitemporal spectral pairs to facilitate discriminative learning under spectral pseudochange. During inference, the predicted pseudoclass label is directly mapped back to the change mask.

bands (marked by the red rectangular box). Ideally, unchanged regions should maintain consistent spectral reflectance at the same wavelength across temporal phases, yet the vertical axis values indicated by red arrows clearly show inconsistent reflectance at identical band positions between different time points. This provides direct visual evidence of pseudochange in multitemporal hyperspectral data. These artifacts can mimic the spectral variation patterns of truly changed pixels, leading to severe distribution overlap between changed and unchanged samples in the bitemporal spectral space. This overlap significantly reduces the separability of the two categories, which is defined by a low interclass distance relative to intraclass variance, making it extremely challenging to accurately distinguish between real change regions and those affected by pseudochange. Therefore, a robust HCD method must effectively enhance the separability between changed and unchanged distributions despite unavoidable radiometric discrepancies, establishing clear decision boundaries to enable precise identification of true surface changes.

Although existing HCD approaches have evolved from traditional difference measures to deep spatial–spectral representation learning, many of them are not specifically designed for pseudochange-induced distribution overlap. Unchanged regions may still produce strong responses due to nuisance spectral discrepancies, and the learned features can be overly sensitive to nondifference patterns, especially under limited supervision.

To address this HSI-specific issue, we adopt a pseudoclass separability perspective on bitemporal spectra and design dedicated spatial–spectral enhancement mechanisms to improve separability under pseudochange. As shown in Fig. 2, given a pixelwise bitemporal spectral pair, we form a concatenated representation and treat changed/unchanged as two pseudoclasses. The use of spectral concatenation is motivated by the fact that elementwise differencing alone preserves only the response magnitude between two temporal phases, but may discard the original spectral context and temporal correspondence needed to distinguish true land-cover changes from radiometric fluctuations. By preserving both prechange and postchange spectra in a unified sequence, the network can jointly observe absolute spectral responses, cross-temporal variations, and peak–valley structural changes before learning discriminative pseudoclass boundaries. It should be emphasized that the term "pseudo-class" does not denote new semantic land-cover categories or pseudolabels generated by unsupervised clustering. At the task level, changed and unchanged regions still correspond to the standard binary change detection labels. We use the term "pseudo-class" to stress that, during training, these labels are treated as discriminative surrogate categories defined on bitemporal spectral pairs. The purpose is not to change the task definition, but to enhance the feature separability of changed and unchanged samples under spectral pseudochange.

Based on this insight, we propose spatial–spectral difference learning (SSDL). First, we introduce a spatial differential attention (SDA) module with a differential mechanism that provides competitive attention adjustment for suppressing common nondiscriminative responses. Second, we develop a spectral attention (SpeA) module that serves as the central spectral-structure enhancement component by modeling peak and valley envelopes with adaptive Gaussian weighting. Third, we further integrate a spectral position encoding (SPE) scheme with position-aware multihead spectral linear self-attention (SpeLA) to model spectral correlations with manageable computational cost.

The order of the three modules follows a coarse-to-fine enhancement rationale: spatial interference suppression, spectral-structure enhancement, and spectral correlation modeling. SDA is first applied to bi-temporal difference features to reduce pseudo-change-induced responses in unchanged regions. SpeA then emphasizes discriminative peak–valley spectral structures and their neighborhoods. SpeLA finally models spectral correlations on the reweighted features, allowing correlation modeling to focus on more informative spectral positions. We do not claim that this ordering is globally opti-

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mal; rather, it follows the above spatial–spectral progressive enhancement rationale.

Unlike methods that mainly prioritize backbone efficiency or generic global context modeling, SSDL directly targets the HSI-specific bottleneck of spectral pseudochange. The pseudoclass separability perspective is used to motivate the design of its enhancement modules, while the main technical emphasis lies in peak–valley spectral enhancement and its integration with spatial differential suppression and efficient spectral correlation modeling. The contributions are summarized as follows.

1) Guided by the pseudoclass separability perspective, we propose SSDL to address the distribution overlap between changed and unchanged bitemporal spectral pairs caused by spectral pseudochange. This perspective provides the design rationale for the spatial–spectral enhancement modules.
2) We propose a peak–valley-guided SpeA module that detects local extrema in the learned spectral feature sequence and applies adaptive Gaussian neighborhood weighting to emphasize discriminative spectral structures.
3) We integrate SDA and position-aware multihead SpeLA into a compact spatial–spectral enhancement framework to suppress nonchange responses and model spectral correlations under pseudochange interference.

The organization of this article is structured as follows. Section II reviews related HCD methods and clarifies the position of our work; Section III elaborates on the proposed methodological framework; Section IV presents the experimental design and results; finally, Section V summarizes the research findings and discusses the limitations of the proposed method.

II. Related Work

A. Difference-Driven and Conventional HCD Methods

Early HCD methods relied on manually designed spectral or spatial features, primarily including spectral matching methods and statistical difference methods. Spectral matching methods identify changes by calculating spectral similarity across temporal phases, with typical examples including the spectral angle mapper [5] and spectral information divergence [6]. These methods assume stable spectral features of unchanged land objects, making them suitable for land types with distinct spectral characteristics. However, they exhibit poor robustness to spectral pseudochanges such as reflectance shifts caused by illumination variations, and often misclassify unchanged regions as changed.

Statistical difference methods extract change regions by computing multitemporal feature differences, typified by change vector analysis [7], its improved Robust change vector analysis (CVA) version [8], and the iteratively reweighted multivariate alteration detection method [9]. These methods are useful for capturing global statistical discrepancies, but they usually depend on hand-crafted difference measures and thresholding strategies. As a result, they struggle to capture fine local changes and cannot sufficiently suppress local interference caused by spectral pseudochange.

B. Deep Spatial–Spectral Representation Learning

With the development of deep learning in computer vision [10], HCD research has shifted toward automatic feature learning via neural networks [11]. Convolutional neural network (CNN)-based methods extract spatial–spectral joint features by stacking 2-D or 3-D convolutional layers. Representative studies include GETNet [12], diverse region-based CNN models [13], and SFBS-FFGNET [14], which improve spatial–spectral feature extraction and alleviate band redundancy to different degrees.

Beyond CNNs, recurrent and graph-based models have also been explored to capture spectral dependencies and spatial neighborhood relationships. For instance, CNN-GNN feature fusion [15], [16], spatial–SpeA ConvLSTM [17], [18], and graph-in-graph convolutional networks [19] strengthen contextual representation learning from different perspectives. Siamese architectures further align features from two temporal phases, such as SSA-SiamNet [20] and GTMSiam [21]. These methods improve the representation ability of HCD models, but they still lack explicit mechanisms for mitigating spectral pseudochange in unchanged regions.

C. Attention-Based and Sequence Modeling Methods

Attention mechanisms have been widely introduced into hyperspectral image analysis and HCD because they can dynamically emphasize informative spatial regions or spectral bands [22]. Spatial–channel attention and cross-attention designs, such as attention-enhanced residual networks [23] and CSCA [24], improve discriminative feature extraction by adaptively weighting spatial–spectral responses. Transformer-based methods further model long-range spatial–spectral dependencies through self-attention [25], including multiattention Transformer models [26] and self-attention networks for hyperspectral image analysis [27].

Recently, state-space-model and Mamba-based architectures [28] have attracted increasing attention due to their ability to model long-range dependencies with improved computational properties. MambaHSI [29] and CDMamba [30] demonstrate the potential of sequence modeling for remote sensing feature extraction. Nevertheless, most existing attention or sequence models mainly focus on global dependence modeling, feature fusion, or computational cost. They do not directly address the HSI-specific distribution overlap between changed and unchanged samples induced by spectral pseudochange.

D. Unsupervised and Limited-Label HCD

Label annotation for HCD is costly because it requires accurate human judgment of land changes across temporal phases. Therefore, unsupervised HCD methods have been investigated to reduce label dependence. Representative methods include unsupervised hierarchical change detection [31], feature mutual representation-based graph domain adaptive networks [32], and binary change-guided hyperspectral multiclass change detection [33]. These methods alleviate reliance

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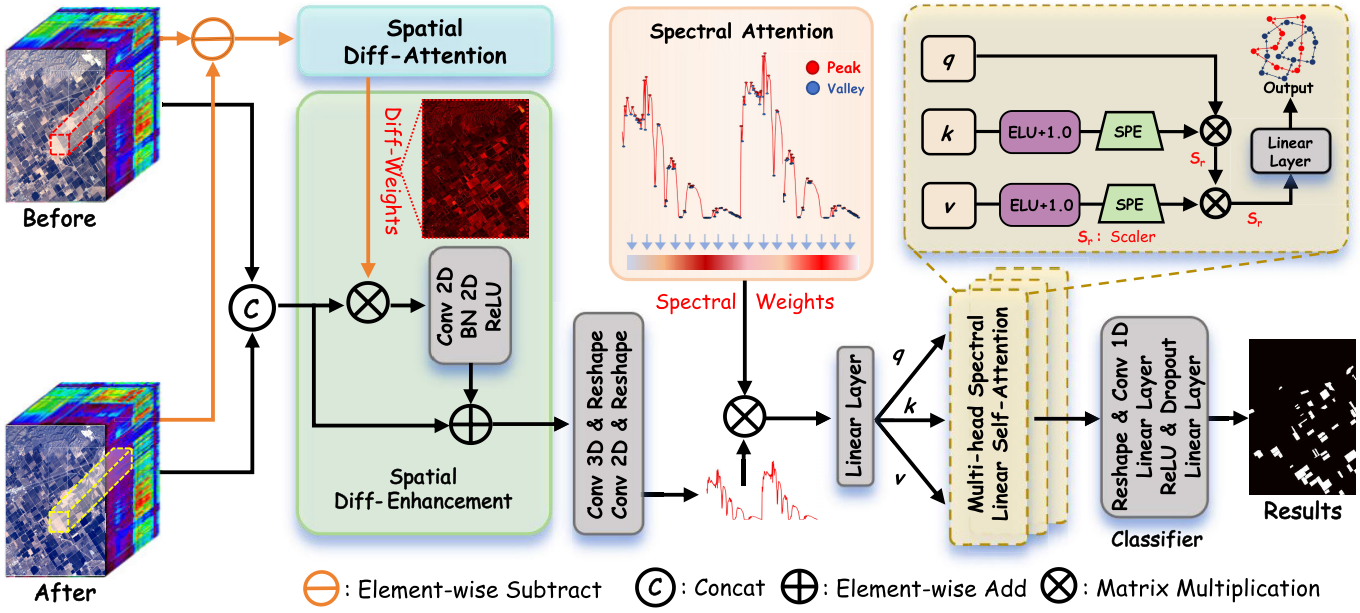


Fig. 3. Overall framework of the SSDL. The proposed SSDL comprises four key components: spatial feature enhancement based on spatial differences, spectral feature enhancement based on peak–valley patterns, SpeLA correlation modeling with position awareness, and a classifier.

on labeled samples, but their learning objectives are not necessarily optimized for suppressing pseudochange-induced false responses under supervised binary HCD settings.

In summary, existing studies have advanced HCD from conventional spectral difference analysis to deep spatial–spectral representation learning, attention-based modeling, and label-efficient learning. However, spectral pseudochange remains a critical bottleneck because unchanged samples may exhibit spectral discrepancies similar to changed samples. This motivates our SSDL framework, which uses pseudoclass separability as a design perspective and introduces dedicated spatial–spectral enhancement modules to improve the discriminability of changed and unchanged bitemporal spectral pairs.

III. Proposed Method

The proposed SSDL sequentially comprises four components: spatial feature enhancement based on spatial differences, spectral feature enhancement based on peak–valley patterns, spectral feature correlation modeling via linear attention, and a classifier for change classification. First, as shown in Fig. 3, cubes are extracted from hyperspectral images of different temporal phases. Next, features from prechange and postchange images undergo concatenation and elementwise subtraction operations. The resulting difference features are fed into our designed SDA module, where the generated difference weights interact with the concatenated features to achieve spatial difference feature enhancement. Subsequently, the enhanced features pass through 3-D convolutional layers, 2-D convolutional layers, and feature transformation to enter the spectral feature enhancement stage. The refined spectral features are then fed into our multihead SpeLA module via linear layers. Finally, a classifier composed of 1-D convolutional layers and linear layers performs change classification, yielding the final change detection results.

A. Spatial Differential Attention

Since changes in land cover types directly alter their spatial features (color and texture), we designed a spatial difference-based attention mechanism. This mechanism highlights spatially distinct changing regions through generated difference weighting for subsequent feature learning. The framework of SDA is shown in Fig. 4. First, the input difference feature D undergoes adaptive average pooling (AAP) along the height and width directions, yielding x_h and x_w. These pooled features are then expanded to $H \times W$ dimensions and summed. After applying sigmoid activation ($\sigma(\cdot)$) to generate spatial weights, the weights are elementwise multiplied with the original feature D, followed by group normalization to complete the initial spatial feature enhancement.

Next, the initially enhanced feature is mapped via a 1×1 convolutional layer to a doubled-channel feature, which is split into dual-path Query (Q = [q_1, q_2]). The original difference feature D is processed through 1×1 convolutional layers: one outputs a doubled-channel feature split into dual-path Key (K = [k_1, k_2]), and the other outputs a single-path Value (V). Subsequently, Q, K, and V are input into the proposed multihead spatial differential self-attention module. Within this module, dual-path attention scores attn1 (dot product of q_1 and k_1) and attn2 (dot product of q_2 and k_2) are computed. Learnable differential parameters (λ_{q1}, λ_{k1}, λ_{q2}, λ_{k2}) are introduced to compute a dynamic coefficient λ (via exponential operations and range clamping), adjusting the attention distribution as (attn_1 - $\lambda \cdot \text{attn}_2$) to suppress sensitivity to nondifference features. The adjusted attention weights are normalized via softmax and applied to V to produce enhanced features. Finally, the enhanced features undergo AAP and sigmoid activation to generate channel weights x_1, while a 3×3 convolution extracts local details

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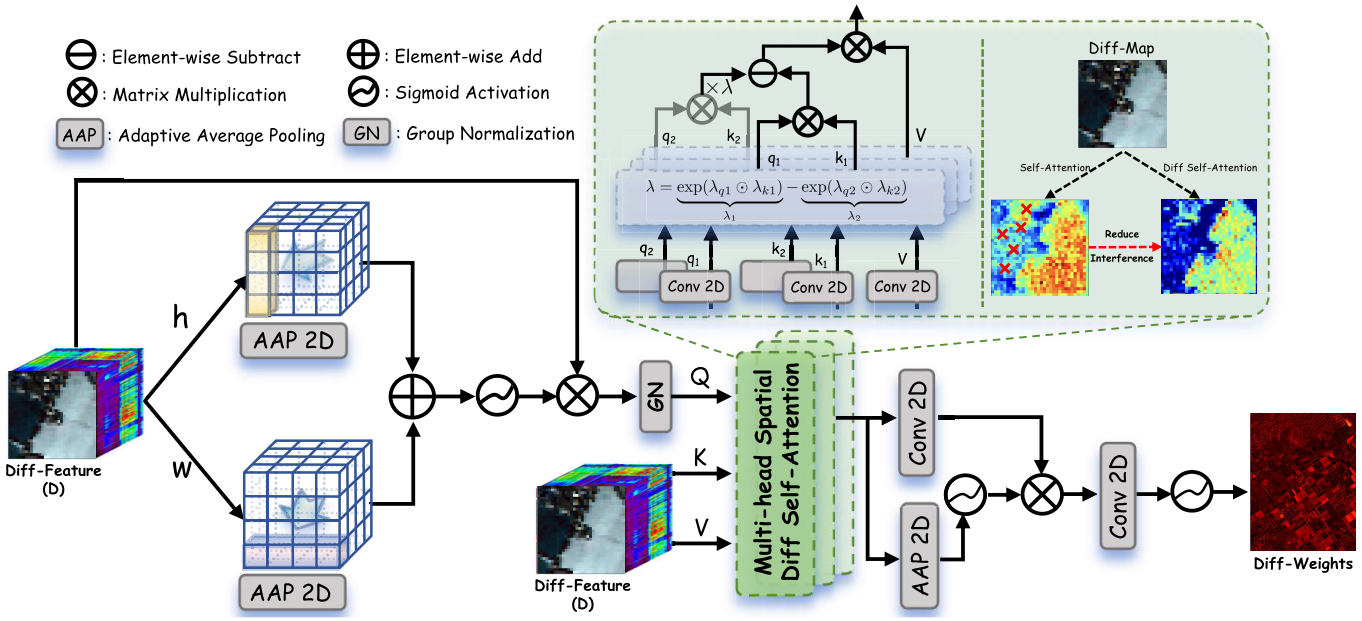


Fig. 4. Workflow framework of the SDA (spatial diff-attention) module. This framework, based on the differential attention mechanism, suppresses the spatial influence of nonchanging regions to highlight the spatial difference features of changing regions.

Algorithm 1 Spatial Differential Attention

Require: Input difference feature map $D \in \mathbb{R}^{B\times C\times H\times W}$
Number of heads h, Head dimension $d = \frac{C}{h}$
Ensure: Spatial attention map $A \in \mathbb{R}^{B\times 1\times H\times W}$
Step 1: Initial Spatial Enhancement
$x_h \leftarrow \text{Reshape}(\text{AAP2D}(D, (H, 1)), [B, C, H, 1])$
$x_w \leftarrow \text{Reshape}(\text{AAP2D}(D, (1, W)), [B, C, 1, W])$
$hw \leftarrow \text{Expand}(x_h, W) + \text{Expand}(x_w, H)$
$D_{\text{enh}} \leftarrow \text{GroupNorm}(D \odot \sigma(hw))$
Step 2: Dual-path Projection
$Q \leftarrow \text{Conv1x1}(D_{\text{enh}}, 2C)$
$K \leftarrow \text{Conv1x1}(D, 2C)$
$V \leftarrow \text{Conv1x1}(D, C)$
Step 3: Head Splitting and Attention Computation
$Q \leftarrow \text{Reshape}(Q, [B, 2, h, d, H \times W])$
$K \leftarrow \text{Reshape}(K, [B, 2, h, d, H \times W])$
$q_1, q_2 \leftarrow Q[:, 0], Q[:, 1],\ k_1, k_2 \leftarrow K[:, 0], K[:, 1]$
$\text{attn}_1 \leftarrow \frac{q_1 k_1^\top}{\sqrt{d}},\ \text{attn}_2 \leftarrow \frac{q_2 k_2^\top}{\sqrt{d}}$
Step 4: Differential Weight Adjustment
$\lambda_1 \leftarrow \exp\left(\sum(\lambda_{q1} \odot \lambda_{k1})\right),\ \lambda_2 \leftarrow \exp\left(\sum(\lambda_{q2} \odot \lambda_{k2})\right)$
$\lambda \leftarrow \text{Clamp}(\lambda_1 - \lambda_2, 0, 1)$
$\text{attn}_{\text{adjusted}} \leftarrow \text{Softmax}(\text{attn}_1 - \lambda \cdot \text{attn}_2)$
Step 5: Feature Aggregation
$V \leftarrow \text{Reshape}(V, [B, h, d, H \times W])$
$F_{\text{attn}} \leftarrow \text{Reshape}(\text{attn}_{\text{adjusted}} \cdot V, [B, C, H, W])$
Step 6: Cross Enhancement
$x_1 \leftarrow \sigma(\text{AAP2D}(F_{\text{attn}}, (1, 1)))$
$x_2 \leftarrow \text{Conv3x3}(F_{\text{attn}})$
$F_{\text{cross}} \leftarrow x_2 \odot x_1$
Output
Diff-Weights $\leftarrow \sigma(\text{Conv1x1}(F_{\text{cross}}))$
return Diff-Weights

x_2. These are elementwise multiplied for cross-enhancement, followed by a 1×1 convolution and sigmoid activation to output the spatial attention map Diff-Weights, thereby further enhancing spatial difference features. In our implementation, 2-D adaptive average pooling (AAP2D) uses output sizes (H, 1) and (1, W) with adaptive stride, and Conv3 $\times$ 3 uses stride = 1 and padding = 1 to preserve spatial size. The algorithmic flow of this process is shown in Algorithm 1.

We introduce a differential mechanism to suppress attention responses caused by nondifference (pseudochange) patterns. Since the two attention branches are not explicitly supervised with separate semantic targets, we do not assign fixed meanings to attn_1 and attn_2 as change-related and nuisance-related correlations. Instead, the dual-branch structure is used to form a competitive attention adjustment. The differential term $\text{attn}_1 - \lambda \cdot \text{attn}_2$ does not rely on a predefined semantic partition between the two branches; rather, it suppresses responses that are consistently activated by the learned attention modes but are weakly discriminative for true changes. The learned coefficient $\lambda \in [0, 1]$, computed from λ_{q1}, λ_{k1}, λ_{q2}, and λ_{k2}, adaptively controls the suppression strength: when λ is small, SDA behaves closer to standard spatial attention, whereas a larger λ strengthens the subtraction of redundant or pseudochange-induced responses. Therefore, SDA reduces false responses in unchanged regions through learned competitive suppression rather than by explicitly separating change and interference branches. In implementation, (λ) is initialized to 0 (thus starting from a standard attention form) and updated via backpropagation; Clamp ensures numerical stability and physical interpretability (subtraction strength bounded).

B. Spectral Attention

Spectral characteristics refer to the response features of ground objects under light of varying wavelengths, with distinct terrestrial object categories exhibiting unique response patterns. These response features often manifest more pronounced discrepancies within the peak–valley regions of spectral curves across different classes. As illustrated in Fig. 5, the spectral features before and after change show

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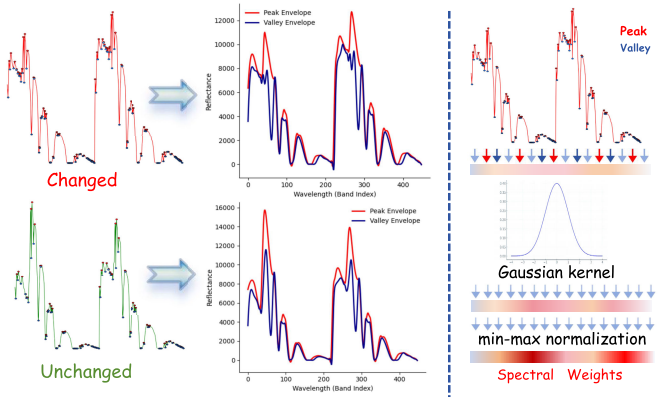


Fig. 5. (Left) Spectral decomposition results based on peak–valley patterns and (right) schematic workflow of our proposed implementation of SpeA.

heightened dissimilarity following peak–valley-based spectral decomposition, making peak–valley features the regions meriting greater attention. Unlike conventional channel attention that learns global bandwise weights, the proposed SpeA explicitly introduces peak–valley positions detected from the learned spectral feature sequence as a spectral-structure prior. Peaks, valleys, and their neighboring bands often correspond to discriminative reflectance or absorption variations, making them more informative than uniformly treating all spectral bands. To avoid the noise sensitivity of hard extrema selection, SpeA uses adaptive Gaussian neighborhood weighting to softly enhance continuous spectral regions around detected peaks and valleys.

SpeA enhances peak–valley regions by processing the learned spectral feature sequence F_s obtained after spatial-difference enhancement and convolutional feature transformation. In particular, for each spectral feature sequence, we compute first-order differences along the spectral-feature dimension and detect turning points according to sign changes. A position is assigned to the peak set P when the difference changes from positive to negative, and to the valley set V when the difference changes from negative to positive. A position grid covering all valid spectral-feature positions is simultaneously generated, and Gaussian weights are computed around the detected peaks and valleys using learnable width parameters. Boundary positions are not treated as strict extrema because they do not have two-sided neighborhoods; however, they can still receive attention weights through truncated Gaussian neighborhoods generated by nearby detected extrema. In particular, the peak weight w_p and valley weight w_v are implemented as learnable parameters in logarithmic space to ensure nonnegative values during training. We initialize $\log(w_p)$ and $\log(w_v)$ with $\log(3)$, and optimize these parameters end-to-end via backpropagation. The exponential transformation $w_p = \exp(\log(w_p))$ and $w_v = \exp(\log(w_v))$ guarantee strictly positive weight values, maintaining numerical stability and physical interpretability of the Gaussian kernel throughout training.

For each detected center ($c \in P \cup V$) (band index), we place a Gaussian kernel ($\mathcal{N}(c, w^2)$) to model a soft neighborhood around the peak/valley. Equation (1) sums peak- and valley-induced weights because both correspond to highly discriminative spectral structures; combining them yields a complete attention profile over salient positions. For flat regions, small first-order differences whose absolute values are below a threshold are treated as zero. Consecutive zero-difference positions are regarded as a plateau: if the signs on the two sides of the plateau form a positive-to-negative transition, the center of the plateau is treated as a peak; if they form a negative-to-positive transition, the center is treated as a valley; otherwise, the plateau is not selected as an extremum. The resulting SpecAtt $\in \mathbb{R}^{B\times C\times N}$ is min–max normalized to [0, 1] and is used to reweight spectral channels (elementwise multiplication along the channel/band dimension) to emphasize peak–valley discrepancies between pseudoclasses.

The process of SpeA can be formally expressed as follows:

$$\mathcal{S}pecAtt = \text{Norm}\left(\sum_{c\in P}\mathcal{N}\left(c, w_p^2\right) + \sum_{c\in V}\mathcal{N}\left(c, w_v^2\right)\right) \quad (1)$$

where $\mathcal{N}(\mu, \sigma^2)$ denotes the Gaussian kernel, and Norm$(\cdot)$ is the min–max normalization.

C. Multihead SpeLA

In SSDL, the position-aware multihead SpeLA module is used as an efficient spectral correlation modeling component after peak–valley spectral enhancement, rather than as the primary source of novelty. Its role is to capture dependencies among reweighted spectral features with manageable computational cost, thereby complementing the local spectral-structure emphasis introduced by SpeA. In hyperspectral image analysis, such correlation modeling helps the network compare the spectrum at a specific position with its surrounding spectral context and refine the discrimination between pseudoclasses.

During initialization, the input feature dimension is denoted as C, and the number of attention heads is set to n, resulting in a per-head feature dimension of $d = C/n$. The spatial resolution of the input feature map is $H \times W$, leading to a total number of spectral feature tokens $N = H \times W$. Here, N is not the number of spectral bands; each token corresponds to a spatially indexed C-dimensional spectral feature embedding.

In operation, the input feature tensor $\mathbf{x} \in \mathbb{R}^{B\times N\times C}$ is first transformed into query, key, and value representations through a linear projection. In particular, a single linear layer maps the input features to a 3C-dimensional space, which is then evenly split along the channel dimension to obtain the query, key, and value tensors. Each of these tensors is further reshaped into multiple attention heads, yielding $\mathbf{q}$, $\mathbf{k}$, $\mathbf{v} \in \mathbb{R}^{B\times n\times N\times d}$. This multihead decomposition allows the model to capture diverse spectral correlations from different representation subspaces.

The algorithmic flow of this process is shown in Algorithm 2. To ensure nonnegative kernel features and improve numerical stability, we adopt ELU$(\cdot)$+ 1.0 for the transformed keys/values, which stabilizes linear attention computation by preventing negative correlations from causing cancellation.

1) Spectral Position Encoding: The process of SPE begins by receiving hyperspectral feature inputs. As shown in Fig. 6, the SPE strengthens the module's ability to perceive position-dependent spectral patterns, defined by the following equation:

$$\mathbf{PE} = \text{G}\left(\text{P}\left(\text{C}\left[\sin\left(\omega_k \cdot (i-j)\right), \cos\left(\omega_k \cdot (i-j)\right)\right]_{k=1}^{d/2}\right)\right) \quad (2)$$

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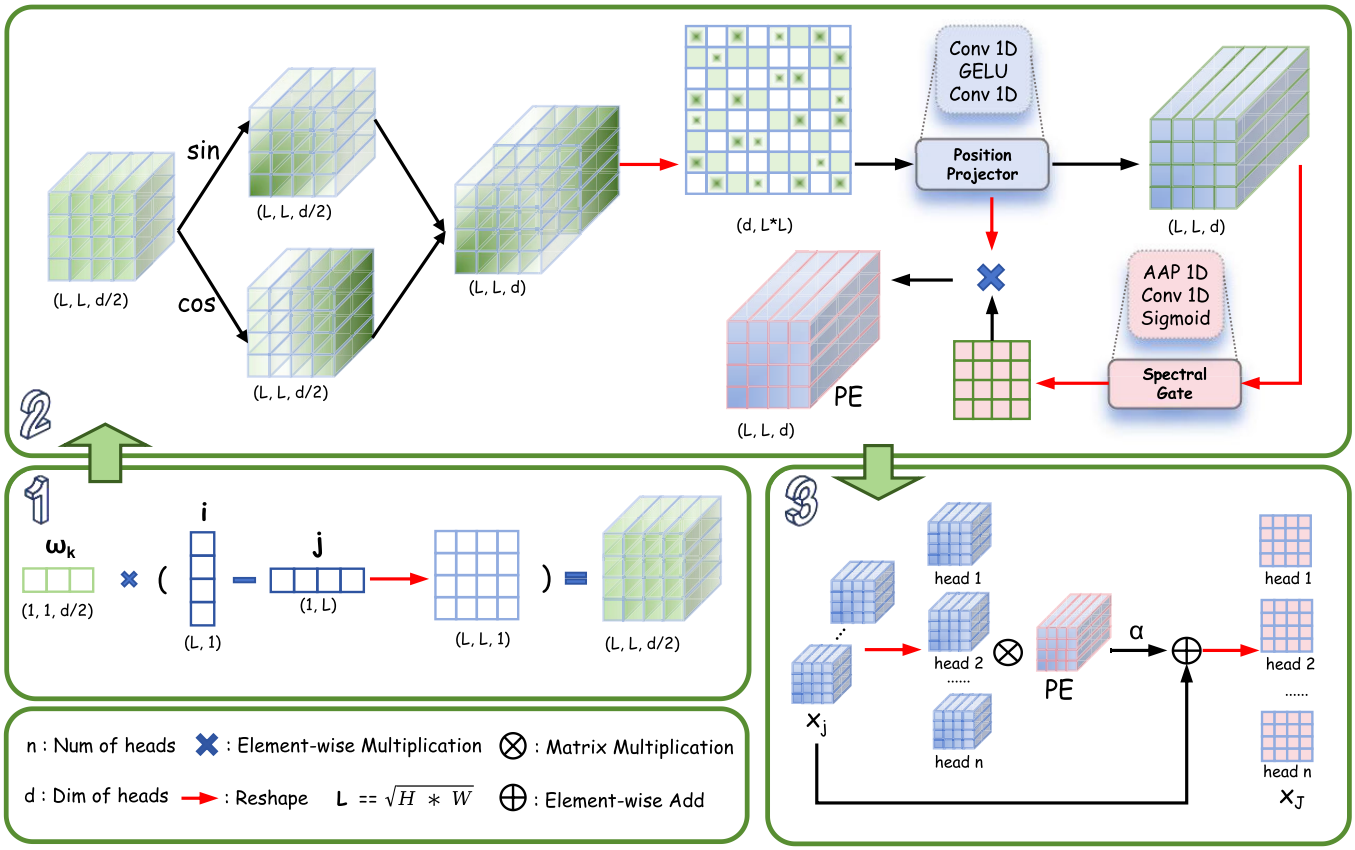


Fig. 6. Flowchart of SPE. After being mapped via a linear layer, keys (k) and values (v) are subjected to exponential linear unit (ELU) activation, then input into the SPE module to establish positional correlations between spectra, thereby enhancing intraclass similarity among the same categories.

Algorithm 2 Multihead SpeLA

Require: $X \in \mathbb{R}^{B\times N\times C}$
Ensure: $Output \in \mathbb{R}^{B\times N\times C}$
$n \leftarrow \text{num_heads},\ d \leftarrow C/n,\ S_r \leftarrow (d\cdot N)^{-0.5}$
$(q,k,v) \leftarrow \text{LinearProj}(X, 3C)$
$A_{\text{attn}} \leftarrow \text{zeros}(B,n,N,d)$
for $j \leftarrow 1$ to n **do**
  $(K_j, V_j) \leftarrow \text{ELU}(k[j]) + 1.0,\ \text{ELU}(v[j]) + 1.0$
  $(K_J, V_J) \leftarrow \text{SPE}(K_j, V_j, (H, W))$
  $C_J \leftarrow S_r \cdot (K_J^{\top} \cdot q[j])$
  $A_{\text{attn}}[j] \leftarrow S_r \cdot (V_J \cdot C_J)$
end for
$Output \leftarrow \text{Linear}(A_{\text{attn}}, C)$
return $Output$

where **PE** represents the SPE at spatial position (i, j) (dimension d) that captures local spectral variations; i, j are spatial indices $(0 \le i, j < L)$ corresponding to 2-D coordinates; ω_k represents the learnable frequency basis (tunable during training) that modulates the encoding's sensitivity to spectral patterns; C(·) represents the concat; P(·) represents the learnable Position Projector (e.g., Conv1d + gaussian error linear unit (GELU) + Conv1d) that enhances encoding expressiveness; and G(·) represents the Spectral Gate module (e.g., adaptive pooling + Conv1d + Sigmoid) that dynamically adjusts encoding importance based on local features.

This involves calculating contextual spectral information from **K** and **V**, then combining it with **Q** and **PE** to generate the final output. The spectral position interaction is formalized as follows:

$$\mathbf{x}_J = \mathbf{x}_j + \alpha \cdot \left(\mathbf{x}_j \otimes \mathbf{PE}\right) \tag{3}$$

where $\mathbf{x}_J$ represents the final output tensor after applying SPE and $\mathbf{x}_j$ represents the original input tensor before applying SPE; $\otimes$ denotes elementwise multiplication, fusing reshaped features with positional encoding to highlight position-dependent spectral correlations; and α represents the scaling factor ($\alpha = d^{-1/4}$) that balances the influence of positional encoding to avoid overamplification.

Instead of using traditional 2-D dot-product attention (which incurs high computational costs), a linear attention mechanism is employed. Multihead SpeLA involves calculating contextual information from the key and value features, then combining it with the query features to generate the final attention output. This efficient process captures the changing patterns and correlations of surrounding spectra. Finally, the outputs from multiple attention heads are aggregated and projected through a transformation layer. In this way, SpeLA models dependencies among informative spectral positions and complements the peak–valley enhancement of SpeA, enabling the model to refine the discrimination of whether spectral changes have occurred at the target position.

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D. Classifier

The enhanced spatial–spectral features are finally fed into a compact classifier for pseudoclass prediction, so that the preceding feature enhancement modules can improve accuracy without excessively increasing the overall complexity. Let the input feature be $\mathbf{F} \in \mathbb{R}^{B\times C\times N}$, where B, C, and N denote the batch size, feature dimension, and sequence length, respectively.

To aggregate discriminative spectral responses, we first apply an attention-based pooling operation implemented by a 1×1 convolution along the feature dimension, as shown by the following equation:

$$\mathbf{a} = \text{squeeze}\left(\text{Conv1D}\left(\mathbf{F}\right)\right), \quad \mathbf{a} \in \mathbb{R}^{B\times N}. \tag{4}$$

This operation adaptively emphasizes informative spectral positions while suppressing irrelevant responses.

The pooled features are then passed through a shallow multilayer perceptron consisting of two fully connected layers with ReLU activation and dropout regularization to produce the final prediction logits, as shown by the following equation:

$$\hat{\mathbf{y}} = \text{MLP}\left(\mathbf{a}\right) \tag{5}$$

where $\hat{\mathbf{y}} \in \mathbb{R}^{B\times K}$ and K is the number of pseudo-classes.

This lightweight classifier effectively aggregates salient spectral cues and preserves the discriminative representations learned by the preceding modules, enabling stable and accurate pseudoclass prediction.

IV. Experiments

To validate that each module proposed in our SSDL can enhance the accuracy and performance of HCD to varying degrees, we designed an extensive series of ablation experiments and comparative experiments. The effectiveness of our approach was further supported through the visualization of experimental results and processes. To ensure fairness, each experiment was repeated ten times, and the final results were presented as the average values and standard deviations for comparison. The models were trained using 1% of the labeled pixels and tested on the remaining 99% to generate all results. The labeled training samples were obtained from the available ground-truth change maps, where changed and unchanged pixels were treated as two pseudoclasses for supervised binary prediction. SSDL was trained end-to-end with cross-entropy loss and did not use additional unlabeled-data objectives, pseudolabel generation, or external pretraining. Accordingly, the conclusions drawn from the experimental results should be interpreted under this fixed 1% labeled-sample setting. The 1% labeled training samples were selected once before the ten repeated runs, and the same training/testing split was reused for all runs. Under this limited-label protocol, all methods use the same training/testing split, the same labeled-sample ratio, and the same evaluation metrics. For baseline methods, we follow their public implementations and recommended configurations whenever available. When exact configurations are not available, we keep the training epochs, optimizer, learning-rate schedule, and input patch settings consistent. The goal is not to perform dataset-specific exhaustive tuning for each baseline, but to compare model robustness under a unified limited-label setting.

A. HCD Datasets

We selected three publicly available and challenging HCD datasets to validate our method, comprising two large-scale datasets (Santa Barbara and Bay Area) and one small-scale dataset (Farmland).

1) Santa Barbara Dataset: The Santa Barbara dataset focuses on the Santa Barbara region in CA, USA, with data collected by the airborne visible/infrared imaging spectrometer (AVIRIS) sensor in 2013 and 2014, respectively. It contains two hyperspectral images of 984×740 pixels, with a spectral resolution of 10 nm, 224 spectral bands, primarily covering typical land features such as mountainous areas, roads, and farmlands.

2) BayArea Dataset: The Bay Area dataset covers the Patterson area and its surroundings in CA, USA, with data acquired by the AVIRIS sensor. It includes bitemporal hyperspectral images captured in 2013 and 2015. The images are 600×500 pixels in size, with a spatial resolution of 20 m, 224 spectral bands, and a wavelength range spanning from 0.4 to 2.5 μm.

3) Farmland Dataset: The Farmland dataset encompasses a farmland region near Yancheng, Jiangsu, China. The data were collected via the Earth Observing-1 (EO-1) Hyperion sensor, capturing bi-temporal hyperspectral imagery in 2006 and 2007. The original imagery contained 242 spectral bands spanning from 0.4 to 2.5 μm, with a 30-m spatial resolution. After excluding noisy and water absorption bands, the dataset was refined to 155 spectral bands. Each image in this set has spatial dimensions of 450×140 pixels, and the primary region of change within the dataset focuses on land cover dynamics.

B. Experimental Setup

1) Evaluation Metrics: We employed five metrics commonly used in HCD tasks: overall accuracy (OA), Kappa coefficient (κ), $F1$ score ($F1$), precision (P), and recall (R). These metrics enable a comprehensive and fair comparison of model performance and the effectiveness of different modules.

2) Hardware Environment: All experiments were conducted on a server equipped with an Intel Xeon[1] Silver 4210R CPU and an NVIDIA Quadro RTX 5000 16 GB graphics processing unit (GPU).

3) Comparative Methods: To demonstrate the superior performance of our proposed SSDL in HCD, we compared it with other competitive methods, including ReCNN [34], GETNet [12], MLEDAN [35], CSANet [36], BIT [37], BTCDNet [38], AIWSEN [39], and CGMNet [40]-all of which are open-sourced in their original papers. During comparisons, we maintained consistent parameter settings with the original networks. All methods were trained for 100 epochs across the three datasets. The optimizer was uniformly configured as stochastic gradient descent (SGD), with an initial learning rate set to $5e^{-3}$ (decayed by a factor of ten every 35 epochs). Meanwhile, the loss function was defined as cross-entropy loss. In addition, the patch sizes for the BayArea, Santa Barbara, and Farmland datasets were set to 14, 14, and 8, respectively.

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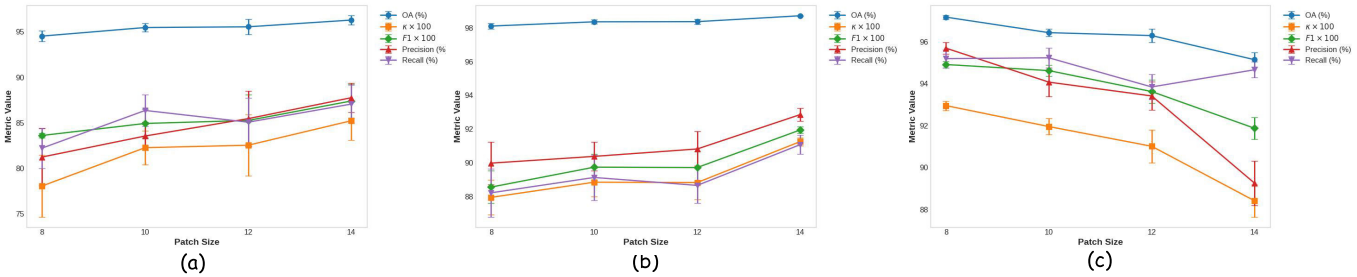


Fig. 7. Comparison of experimental results with different patch sizes across three datasets. (a) BayArea dataset. (b) Santa Barbara dataset. (c) Farmland dataset.

TABLE I

COMPARISON OF RESULTS USING DIFFERENT SPATIAL ATTENTION MECHANISMS ON THE BAYAREA DATASET. BEST RESULTS ARE IN BOLD

Metric	NL2D	CBAM	EMA	CSCA	SDA(Ours)
OA (%)	94.97±1.61	95.83±0.06	96.05±0.10	95.88±0.10	**96.30±0.53**
κ*100	80.13±6.50	83.19±0.30	84.16±0.39	83.75±0.40	**85.24±2.14**
F1*100	83.08±5.56	85.62±0.27	86.47±0.33	86.17±0.35	**87.41±1.84**
Precision (%)	82.34±5.01	87.38±0.42	87.54±0.54	85.50±0.59	**87.77±1.62**
Recall (%)	83.86±6.81	83.94±0.08	85.43±0.55	86.86±0.72	**87.07±2.08**

TABLE II

COMPARISON OF RESULTS USING DIFFERENT LINEAR ATTENTION MECHANISMS ON THE BAYAREA DATASET. BEST RESULTS ARE IN BOLD

Metric	Eff-Att	Agent-Att	MLLA	PolaLA	SpeLA(Ours)
OA (%)	95.74±1.15	95.88±0.20	95.76±0.08	95.28±0.81	**96.30±0.53**
κ*100	82.39±5.10	83.56±0.89	83.17±0.34	80.77±3.93	**85.24±2.14**
F1*100	84.86±4.45	85.98±0.77	85.65±0.29	83.51±3.50	**87.41±1.84**
Precision (%)	**87.82±3.03**	86.41±0.48	85.64±0.37	85.81±1.05	87.77±1.62
Recall (%)	81.30±5.68	85.56±1.32	85.67±0.51	81.48±5.42	**87.07±2.08**

4) Parameter Setting of Patch Size: To investigate the input feature size requirements of the proposed method, we set different patch sizes across the three datasets: specifically, the patch sizes for the three datasets were configured as [8, 10, 12, 14] (where a patch size exceeding 14 would exceed GPU memory capacity in Bay Area and Santa Barbara). The patch size controls the amount of spatial context used for each target pixel. A small patch may provide insufficient neighborhood information, whereas an overly large patch may introduce irrelevant background pixels, increase computational cost, and aggravate overfitting under the fixed 1% labeled-sample setting. Therefore, we evaluate several candidate patch sizes, i.e., 8×8, 10×10, 12×12, and 14×14, under the same training protocol. The final patch size is selected according to the sensitivity analysis in Fig. 7. In particular, 8×8 provides a compact local neighborhood and is more suitable for the smaller Farmland dataset, while larger patches benefit BayArea and Santa Barbara by providing richer spatial context when sufficient samples are available.

Experimental results (as shown in Fig. 7) indicate that the performance of our method improves with increasing patch size on BayArea and Santa Barbara, yet on Farmland, optimal performance is achieved when the patch size is set to 8. Furthermore, analysis suggests that the selection of patch size is closely related to the number of training samples—when the training sample size is sufficiently large, the method's performance can be further enhanced with larger patch sizes (potentially even outperforming the results reported in this article on BayArea and Santa Barbara). This characteristic better aligns with the demands of future large-scale datasets.

C. Comparison Experiment

1) Comparison With Spatial Attention Variants: To evaluate the effectiveness of our proposed SDA in extracting discriminative spatial change features, we conducted comparative experiments on the BayArea dataset between SDA and competing spatial attention mechanisms, including NonLocal Attention (NL2D) [41], CBAM [42], EMA [43], and CSCA [24]. In particular, since none of the compared methods alter feature tensor dimensions, we directly fed temporal difference features into different attention modules while keeping all other network components identical. As shown in Table I, SDA achieves the best performance across all evaluation metrics in this controlled comparison, indicating its suitability for feature change highlighting.

2) Comparison With Linear Attention Variants: To evaluate the effect of our multihead SpeLA relative to other linear attention methods, we compared SpeLA with EfficientAttention (Eff-Att) [44], AgentAttention (AgentAtt) [45], MLLA [46], and PolaLinearAttention (PolaLA) [47] on the BayArea dataset. We directly replaced SpeLA with these alternative linear attention modules while maintaining identical configurations for other network components. The experimental results (as shown in Table II) indicate that SpeLA achieves better results in four metrics (excluding P), confirming that modeling spectral correlations has a positive impact on subsequent change classification performance.

3) Comparison With Existing HCD Methods: Table III illustrates the comparison of the proposed SSDL with eight representative HCD methods. Among the compared methods, SSDL achieves the best results across all five evaluation metrics on both the BayArea and Santa Barbara datasets. Compared with the strongest competing result for each metric, SSDL improves OA, κ, $F1$, precision, and pecall by 1.30, 5.23, 4.75, 5.32, and 6.42 points, respectively, on BayArea, and by 0.35, 2.23, 2.14, 0.59, and 3.30 points, respectively, on Santa Barbara. On the Farmland dataset, SSDL ranks first in OA, κ, $F1$, and precision, exceeding the corresponding strongest baselines by 0.28, 0.43, 0.18, and 1.54 points, respectively. Although its Recall (95.19%) is lower than the best competing result (96.36%), its leading precision and $F1$ scores indicate a more favorable balance between missed detections and false

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TABLE III

CHANGE DETECTION RESULTS OBTAINED WITH DIFFERENT NETWORKS ACROSS THREE DATASETS. BEST RESULTS ARE IN BOLD

Method	BayArea			Santa Barbara			Farmland		
	OA (%)	κ×100	F1×100	OA (%)	κ×100	F1×100	OA (%)	κ×100	F1×100
ReCNN	86.13±1.09	15.24±14.83	18.12±17.02	97.20±0.08	79.56±0.64	81.06±0.60	93.36±0.75	84.47±1.86	89.24±1.34
GETNet	92.73±0.17	70.40±0.64	74.63±0.54	97.55±0.10	82.26±0.86	83.57±0.81	96.16±0.08	90.80±0.19	93.52±0.12
MLEDAN	89.67±1.99	47.87±18.43	52.45±19.42	97.87±0.02	85.08±0.18	86.23±0.17	96.02±0.34	91.88±0.38	93.33±0.37
CSANet	95.00±0.32	80.01±1.54	82.66±1.36	98.35±0.04	89.03±0.32	89.81±0.30	96.32±0.19	91.12±0.44	93.72±0.30
BIT	93.84±0.17	74.72±0.80	78.30±0.71	98.37±0.03	88.71±0.20	89.59±0.20	96.03±0.15	90.51±0.37	93.33±0.26
BTCDNet	89.87±0.20	52.40±1.04	57.85±0.83	95.03±0.17	55.27±1.81	57.59±1.81	96.59±0.17	91.81±0.93	94.22±0.28
AIWSEN	92.00±0.13	67.17±0.88	71.81±0.83	97.42±0.02	82.13±0.17	83.53±0.17	96.89±0.06	92.52±0.16	94.73±0.11
CGMNet	93.06±0.14	71.81±0.70	75.85±0.63	97.75±0.18	84.49±1.26	85.71±1.16	96.41±0.08	91.39±0.18	93.95±0.13
SSDL(Ours)	**96.30±0.53**	**85.24±2.14**	**87.41±1.84**	**98.72±0.03**	**91.26±0.23**	**91.95±0.20**	**97.17±0.09**	**92.95±0.22**	**94.91±0.16**
Method	BayArea			Santa Barbara			Farmland		
	Precision (%)	Recall(%)	-	Precision (%)	Recall(%)	-	Precision (%)	Recall(%)	-
ReCNN	48.32±33.50	12.29±12.70	-	88.87±0.47	74.52±0.85	-	84.12±0.91	95.05±2.34	-
GETNet	77.10±0.97	72.33±0.79	-	90.93±0.73	77.34±1.65	-	91.61±0.08	95.52±0.38	-
MLEDAN	76.00±3.18	43.14±17.86	-	90.03±0.11	82.74±0.26	-	93.03±0.21	95.25±0.36	-
CSANet	82.45±0.67	80.65±2.36	-	90.95±0.30	87.78±0.63	-	92.96±0.79	94.50±0.47	-
BIT	81.63±0.90	75.26±1.48	-	92.27±0.40	87.06±0.50	-	91.06±0.42	95.72±0.42	-
BTCDNet	74.02±1.42	47.50±1.29	-	86.67±1.31	43.15±1.86	-	92.75±0.68	95.75±0.54	-
AIWSEN	74.92±0.71	68.99±1.89	-	86.10±0.67	81.11±0.81	-	94.15±0.17	**96.36±0.16**	-
CGMNet	78.08±0.58	73.76±1.28	-	87.87±1.24	83.66±1.36	-	91.84±0.50	96.16±0.59	-
SSDL(Ours)	**87.77±1.62**	**87.07±2.08**	-	**92.86±0.38**	**91.08±0.57**	-	**95.69±0.27**	95.19±0.43	-

TABLE IV

ABLATION RESULTS OF USING SDA ON THE BAYAREA AND SANTA BARBARA DATASET. BEST RESULTS ARE IN BOLD

Metric	BayArea			Santa Barbara		
	No_SDA	No_Diff	Ours	No_SDA	No_Diff	Ours
OA (%)	95.35±1.17	95.91±0.08	**96.30±0.53**	98.57±0.50	98.62±0.49	**98.72±0.03**
κ*100	81.64±4.76	83.88±0.32	**85.24±2.14**	90.54±3.31	90.64±3.18	**91.26±0.23**
F1*100	84.37±4.08	86.28±0.28	**87.41±1.84**	91.31±3.04	91.39±2.92	**91.95±0.20**
Precision (%)	83.64±3.52	85.59±0.65	**87.77±1.62**	89.42±2.89	91.79±3.57	**92.86±0.38**
Recall (%)	85.13±4.70	87.00±0.89	**87.07±2.08**	90.29±3.24	91.02±2.65	**91.08±0.57**

TABLE V

ABLATION RESULTS OF USING SPEA ON THREE DATASETS. BEST RESULTS ARE IN BOLD

Metric	BayArea		Santa Barbara		Farmland	
	No_SpeA	Ours	No_SpeA	Ours	No_SpeA	Ours
OA (%)	95.41±1.08	**96.30±0.53**	98.52±0.14	**98.72±0.03**	96.48±0.23	**97.17±0.09**
κ*100	81.98±4.19	**85.24±2.14**	90.85±0.33	**91.26±0.23**	91.45±0.55	**92.95±0.22**
F1*100	84.68±3.55	**87.41±1.84**	91.35±0.27	**91.95±0.20**	93.93±0.39	**94.91±0.16**
Precision (%)	83.52±3.64	**87.77±1.62**	91.79±0.57	**92.86±0.38**	93.94±0.69	**95.69±0.27**
Recall (%)	85.88±3.53	**87.07±2.08**	90.62±0.65	**91.08±0.57**	93.92±0.74	**95.19±0.43**

alarms. These results demonstrate the robustness and competitiveness of SSDL across scenes of different scales. Together with the ablation results and qualitative feature visualizations, the comparison results also support the intended role of SSDL in reducing pseudochange-induced false responses, although the comparative results alone should not be interpreted as direct proof of the underlying mechanism.

D. Ablation Study

1) Validation of SDA Module and Differential Mechanism: To verify the effectiveness of our designed SDA module and differential mechanism, we conducted experiments on the BayArea and Santa Barbara datasets without the SDA module (No_SDA) and with the SDA module but without the differential mechanism (No_Diff). Here, the differential mechanism refers to the dual-path attention with dynamic differential coefficient (λ), where the attention logits are adjusted as $(attn_1 - \lambda \cdot attn_2)$ to suppress shared nondiscriminative responses. In "No_Diff," we remove the second path and the subtraction term, reducing SDA to a standard spatial attention enhancement. The results are presented in Table IV. No_Diff consistently improves over No_SDA, indicating that spatial attention enhancement itself is beneficial for extracting difference-aware features. More importantly, Ours further outperforms No_Diff on both datasets, e.g., improving the $F1$ score from 86.28 to 87.41 on BayArea and from 91.39 to 91.95 on Santa Barbara. This additional gain supports the role of the differential term and the learned coefficient λ: the subtraction term provides competitive suppression beyond standard spatial attention, while λ controls the degree of this suppression during training. Therefore, the ablation results suggest that SDA suppresses pseudo-change-induced responses in unchanged regions without requiring the two attention branches to be explicitly supervised as change and interference branches. This result also suggests that SDA contributes to suppressing misresponses in unchanged areas. Compared with No_Diff, the full SDA introduces competitive subtraction between two attention branches, which weakens common responses that are not strongly related to real changes. Therefore, the gain of Ours over No_Diff can be interpreted as evidence that the differential mechanism helps reduce pseudochange-induced false responses.

2) Validation of SpeA: To assess the efficacy of our designed SpeA, we conduct experiments without SpeA on all three datasets, with results shown in Table V. The findings demonstrate that the performance of experiments using SpeA (Ours) surpasses that of those without SpeA (No_SpeA) across all datasets. Experimental analysis indicates that enhancing features at spectral peak-trough positions exerts a positive impact on distinguishing between pseudoclass spectra, thereby increasing interclass discriminability. From the perspective of spectral pseudochange suppression, SpeA reduces the influence of unstable but less discriminative

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TABLE VI
ABLATION RESULTS OF USING SPELA ON THE BAYAREA AND FARMLAND DATASET. BEST RESULTS ARE IN BOLD

Metric	BayArea			Farmland		
	No_SpeLA	No_SPE	Ours	No_SpeLA	No_SPE	Ours
OA (%)	95.56±0.32	95.88±0.09	**96.30±0.53**	96.02±0.07	96.61±0.27	**97.17±0.09**
κ*100	83.20±0.36	83.60±0.40	**85.24±2.14**	90.29±1.89	91.74±0.63	**92.95±0.22**
F1*100	85.76±0.56	86.01±0.35	**87.41±1.84**	93.08±1.44	94.12±0.43	**94.91±0.16**
Precision (%)	85.85±0.35	86.19±0.27	**87.77±1.62**	93.53±0.84	94.68±1.19	**95.69±0.27**
Recall (%)	85.12±0.85	85.85±0.77	**87.07±2.08**	92.74±3.36	93.59±0.63	**95.19±0.43**

spectral fluctuations by assigning higher weights to peak–valley neighborhoods in the learned spectral feature sequence. Modeling these spectral structures helps distinguish true change-related spectral variations from pseudochange responses in unchanged samples.

3) Validation of SpeLA: To validate our proposed SpeLA mechanism, we conducted comparative experiments on the BayArea and Farmland datasets under two conditions: without SpeLA (No_SpeLA) and without positional encoding for spectral feature tokens (No_SPE), as illustrated in Table VI. Observations from the tabulated results show that our spectral self-attention mechanism with positional encoding (Ours) demonstrates the top performance. Furthermore, analysis reveals that modeling correlations between spectral bands is critical for constructing intraclass correlations, particularly exerting a significant influence on the final change classification outcomes. Furthermore, analysis shows that modeling correlations among spectral feature tokens is important for reducing residual ambiguity after SpeA. Compared with No_SpeLA, the full model achieves better performance, indicating that SpeLA helps refine the reweighted features by exploiting contextual correlations among spectral feature tokens. The comparison with No_SPE further suggests that positional information benefits the suppression of ambiguous unchanged-region responses.

E. Visualization Analysis

1) Validation of Differential Attention Mechanism on Spatial Feature Differences: To verify the impact of the differential attention mechanism on extracting spatial feature differences, we conducted visual validation on the BayArea dataset. As shown in Fig. 8(a) and (b) represents the features before and after changes, respectively. Fig. 8(c) illustrates the difference features obtained via elementwise subtraction. Fig. 8(d) displays the feature heatmap visualization without differential attention, while Fig. 8(e) presents a more accurate spatial difference heatmap adjusted by the differential mechanism. Through direct visual comparison, we observe that the method incorporating the differential mechanism not only effectively highlights features in change regions but also suppresses feature learning in nonchange regions, thereby achieving more precise enhancement of features in difference regions. The heatmap visualization in Fig. 8 provides a qualitative observation that SDA suppresses background responses in unchanged regions while maintaining stronger responses in changed areas.

2) Visual Validation of Feature Separation Effectiveness: To intuitively demonstrate the effectiveness of our proposed SSDL framework in feature processing across different stages, we sequentially visualized feature distributions using t-SNE

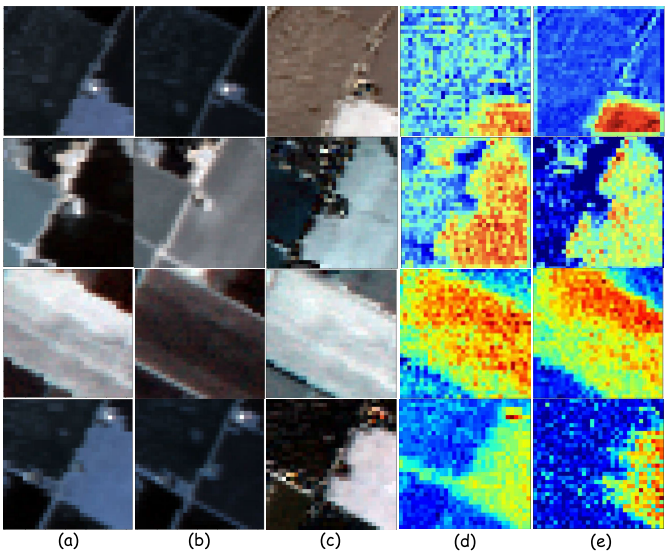


Fig. 8. Visualization of feature heatmaps before and after applying the difference mechanism on the BayArea dataset. (a) Prechange features. (b) Postchange features. (c) Difference features between prechange and postchange features. (d) Heatmaps before using the difference mechanism. (e) Heatmaps after using the difference mechanism.

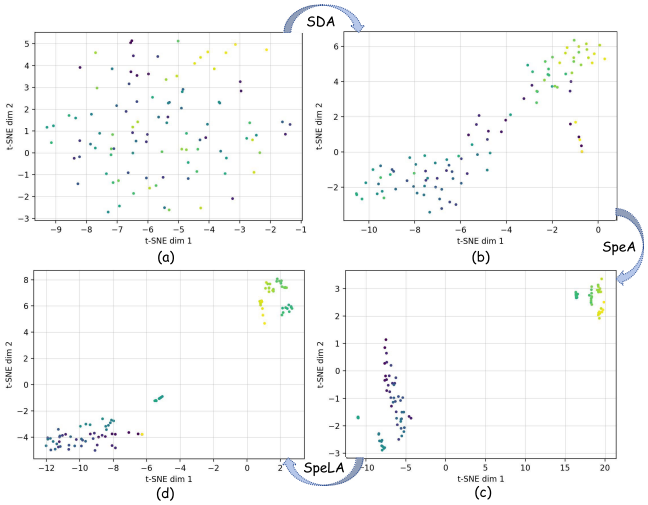


Fig. 9. Visualization of t-SNE after applying the proposed modules on the Barbara dataset. Colors indicate the changed and unchanged pseudo-class labels used for feature visualization. (a) Original features. (b) After applying SDA. (c) After applying SpeA. (d) After applying SpeLA.

on the Santa Barbara dataset by applying our SDA, SpeA, and SpeLA modules respectively, with the results presented in Fig. 9. In particular, we can observe that the raw features without any prior processing exhibit a scattered and irregular distribution pattern [Fig. 9(a)]. After applying SDA, features from changed and unchanged regions become more separable [Fig. 9(b)]. Subsequently, with the enhancement of SpeA, the feature distance between different pseudoclasses is further widened by leveraging the peak–valley characteristics in spectral information, which strengthens the discriminative boundaries between classes [Fig. 9(c)]. Finally, through the correlation modeling learned by SpeLA, samples with the same pseudoclass labels become more compact, while samples with different pseudoclass labels show less overlap in the

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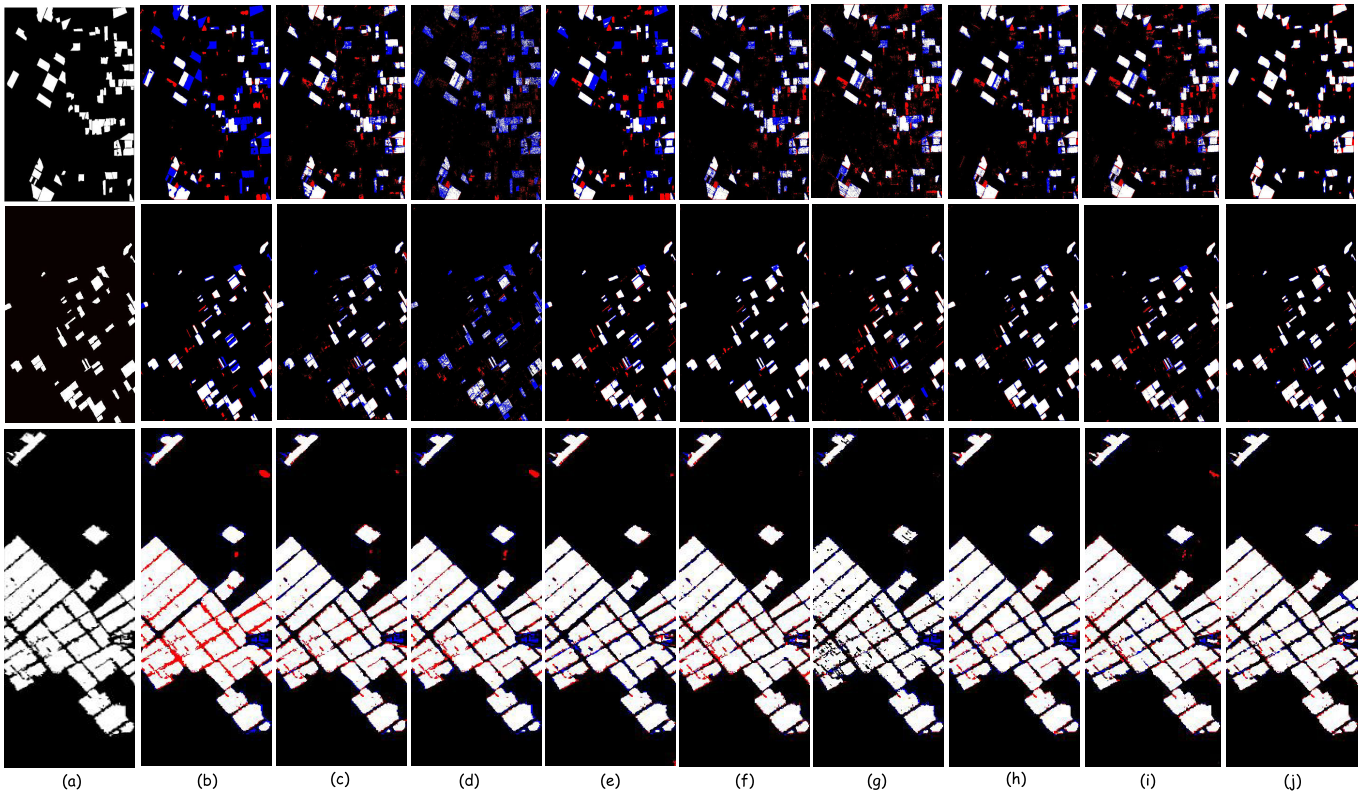


Fig. 10. Visualization of different HCD networks on three datasets. (Top to bottom) BayArea dataset, the Santa Barbara dataset, and the Farmland dataset, respectively. (a) Ground truth. (b) ReCNN [34]. (c) GETNet [12]. (d) MLEDAN [35]. (e) CSANet [36]. (f) BIT [37] (g) BTCDNet [38]. (h) AIWSEN [39]. (i) CGMNet [40]. (j) Proposed SSDL. The FN and FP pixels are denoted in red and blue, respectively.

TABLE VII
COMPARISON OF EFFICIENCY USING DIFFERENT MODELS ON THE FARMLAND DATASET

Metric	ReCNN	GETNet	MLEDAN	CSANet	BIT	BTCDNet	AIWSEN	CGMNet	SSDL(Ours)
Inference Time (ms)	1.7043	2.6499	82.9022	5.9288	41.8115	15.5052	32.0984	22.2916	11.8016
Total Parameters (M)	0.6031	0.2904	112.2859	40.4266	12.6796	1.5649	12.2330	15.3058	1.5930
FLOPs (MFLOPs)	20.6016	16.6723	63.0790	832.9961	48.7135	146.5318	55.9764	349.1306	83.8011

embedded space [Fig. 9(d)]. Compared with Fig. 9(c), the distribution of different pseudoclass samples becomes more distinguishable, with fewer mixed regions and reduced local overlap between changed and unchanged samples. This indicates that SpeLA further refines the feature representation by modeling spectral correlations, enabling the network to better separate samples with similar spectral responses but different change states. Overall, the t-SNE visualization in Fig. 9 qualitatively shows that SDA, SpeA, and SpeLA progressively improve the separation between changed and unchanged pseudoclasses.

3) Visualization of Change Detection Results: We visualize the change detection maps of different HCD methods on the BayArea, Santa Barbara, and Farmland datasets in Fig. 10. In the visualization, red and blue pixels denote false negatives (FNs) and false positives (FPs), respectively. Compared with the baseline methods, the proposed SSDL produces change maps that are generally closer to the ground truth, with fewer visually apparent omission and false-alarm regions. On the BayArea dataset, several competing methods introduce scattered FP responses in unchanged areas and miss parts of compact changed regions, whereas SSDL better preserves the main changed structures while reducing noisy responses. On the Santa Barbara dataset, SSDL also yields cleaner local change boundaries and suppresses many isolated pseudochange responses. For the Farmland dataset, although some boundary-level errors still remain, SSDL maintains more complete farmland parcel structures and shows fewer large-area misclassifications than most comparison methods. Overall, these qualitative results indicate that the proposed SSDL strategy can improve the localization of real changes and reduce pseudochange-induced errors, providing visual support for the effectiveness of SSDL.

F. Efficiency Analysis

To further evaluate the computational characteristics of the proposed method, we compare the inference time, total parameters, and FLOPs of different models on the Farmland dataset, with the results presented in Table VII. The results show that SSDL does not pursue the minimum computational cost in

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terms of parameter number or FLOPs. In particular, SSDL requires 1.5930M parameters and 83.8011 million floating-point operations (MFLOPs), which are higher than lightweight convolutional models such as ReCNN and GETNet, but substantially lower than several attention-based methods, such as CSANet and CGMNet.

Although SSDL introduces additional SDA, peak–valley spectral enhancement, and spectral correlation modeling, it maintains a relatively low inference time of 11.8016 ms, outperforming most compared attention-based approaches. This indicates that the proposed spectral–spatial enhancement modules improve discriminative feature learning without introducing excessive computational overhead. Therefore, the advantage of SSDL lies not in achieving the lowest complexity among all methods, but in achieving a favorable tradeoff between detection performance and computational cost.

V. Conclusion

To mitigate spectral pseudochanges caused by radiometric discrepancies in HCD, we propose a SSDL framework. SSDL does not change the binary decision nature of change detection. Instead, it treats changed and unchanged labels as discriminative surrogate categories defined on bitemporal spectral pairs to enhance feature separability under spectral pseudochange, enabling direct extraction of change maps from classification results.

To enhance feature discriminability while maintaining a favorable accuracy–complexity tradeoff, SSDL incorporates several efficient modules, including SDA for suppressing unchanged regions, peak–valley-based SpeA for emphasizing informative spectral features, and a position-aware multihead SpeLA mechanism for modeling spectral correlations. Extensive experiments on multiple hyperspectral datasets demonstrate that SSDL achieves competitive performance and effectively alleviates spectral pseudochanges.

Although SSDL shows competitive performance, its advantages are more evident on larger scale datasets with sufficient training samples. On small-scale datasets, performance gains are relatively limited, suggesting that future improvements may be achieved by incorporating semisupervised/self-supervised pretraining or pseudolabel refinement under extremely low-data regimes.

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